Project Source Code Reference

AI-Augmented SOC Implementation for
Enterprise Threat Detection and Response

All commands, configurations, and SPL queries
used in the course-end project

1. Splunk Installation and Management (Kali Linux)

1.1 Start Splunk

```
sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root
```

1.2 Check Splunk Status

```
sudo /opt/splunk/bin/splunk status
```

1.3 Stop Splunk

```
sudo /opt/splunk/bin/splunk stop --accept-license --answer-yes --run-as-root
```

1.4 Fix Expired License (Free License)

```
sudo /opt/splunk/bin/splunk stop --accept-license --answer-yes --run-as-root

sudo tee /opt/splunk/etc/system/local/server.conf > /dev/null << 'EOF'
[license]
active_group = SplunkFree
EOF

sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root
```

1.5 Reset Admin Password

```
sudo /opt/splunk/bin/splunk stop --accept-license --answer-yes --run-as-root
sudo rm -f /opt/splunk/etc/passwd
sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --run-as-root
```

> After restart, login with admin / changeme and set a new password.

1.6 Check Listening Ports

```
sudo ss -tlnp | grep 8000
sudo ss -uln | grep 514
```

2. Network Configuration and Verification

2.1 Check Kali IP Address

```
ip addr show | grep "inet "
```

2.2 Ping Test from Metasploitable2 to Kali

```
ping -c 3 192.168.140.128
```

2.3 Netcat Network Test (UDP 514)

On Kali (listener):

```
sudo nc -u -l -p 514
```

On Metasploitable2 (sender):

```
echo "TEST MESSAGE" | nc -u 192.168.140.128 514
```

3. Metasploitable2 Syslog Configuration

3.1 Edit Syslog Configuration

```
sudo nano /etc/syslog.conf
```

> Add this line at the bottom (use Tab, not spaces, between *.* and @host): `*.*  @192.168.140.128`

3.2 Restart Syslog Daemon

```
sudo killall syslogd
sudo syslogd -r
```

3.3 Verify Syslogd is Running

```
ps aux | grep syslogd
```

3.4 Generate Test Log Entry

```
logger "SPLUNK TEST LOG ENTRY"
sudo tail /var/log/auth.log
sudo whoami
```

4. Reconnaissance and Attack Simulation (Kali)

4.1 Nmap ACK Scan (Firewall Mapping)

```
nmap -sA --top-ports 2 192.168.140.131
```

4.2 Nmap SYN Scan (Port Discovery)

```
sudo nmap -sS -p 23,80 192.168.140.131
```

4.3 Nmap Service Version Detection

```
sudo nmap -sV -p 23,80 192.168.140.131
```

4.4 Hydra SSH Brute Force (Attempt - Failed Due to Crypto Mismatch)

```
hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt -t 4 ssh://192.168.140.131
```

4.5 Hydra FTP Brute Force (Successful Attack Vector)

```
hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt -t 4 ftp://192.168.140.131
```

4.6 Unzip Rockyou Wordlist

```
sudo gunzip /usr/share/wordlists/rockyou.txt.gz
ls -la /usr/share/wordlists/rockyou.txt
```

4.7 Hping3 Fragmentation Test

```
sudo hping3 -S 192.168.140.131 -p 80
sudo hping3 -S 192.168.140.131 -p 80 -f
```

5. Splunk SPL Detection Queries

5.1 Verify Log Ingestion

```
index=main sourcetype=syslog | head 20
```

> Time range: All time or Last 15 minutes

5.2 Filter Authentication Failures

```
index=main sourcetype=syslog host=192.168.140.131 "authentication failure"
```

5.3 Search for Specific Username

```
index=main sourcetype=syslog host=192.168.140.131 msfadmin
```

5.4 Statistical Anomaly Detection with Threat Level

```
index=main sourcetype=syslog host=192.168.140.131 "authentication failure"
| rex field=_raw "rhost=(?<attacker_ip>\S+)"
| stats count by attacker_ip
| where count > 0
| eval threat_level=case(count<5,"Low", count<20,"Medium", count⩾20,"High")
| sort - count
```

5.5 IOC Pivot and Timeline Reconstruction

```
index=main sourcetype=syslog host=192.168.140.131 192.168.140.128
| sort + _time
| table _time, _raw
```

5.6 Check Splunk Internal TCP Input Logs

```
index=_internal source=*splunkd.log* tcpin
```

6. Simulated EDR: Linux Auditd (Metasploitable2)

6.1 Install and Configure Auditd

```
sudo apt-get install auditd
sudo auditctl -w /etc/passwd -p wa -k identity_changes
sudo auditctl -w /var/log/auth.log -p wa -k auth_access
```

6.2 Search Auditd Logs in Splunk

```
index=main sourcetype=syslog ("auditd" OR "audit" OR "pam_unix")
| stats count by host, _raw
```

7. Simulated Threat Intelligence: Splunk Lookup

7.1 Create Threat Intel CSV

```
sudo nano /opt/splunk/etc/apps/search/lookups/threat_intel.csv
```

> Contents: ip_address,threat_actor,malware_family,confidence
> 192.168.140.128,Simulated_Attacker_Node,BruteForce_Tool,High

7.2 Enriched Detection Query with Lookup

```
index=main sourcetype=syslog host=192.168.140.131 "authentication failure"
| rex field=_raw "rhost=(?<attacker_ip>\S+)"
| lookup threat_intel ip_address AS attacker_ip OUTPUT threat_actor, malware_family,
confidence
| stats count by attacker_ip, threat_actor, malware_family, confidence
```

8. Simulated SOAR: Splunk Alert Configuration

8.1 Save Detection Query as Alert

In Splunk web interface:

1. Run the anomaly detection query (Section 5.4)
2. Click **Save As** → **Alert**
3. Title: `FTP_Brute_Force_Detected`
4. Alert type: **Scheduled** (every 5 minutes)
5. Trigger condition: **Number of results > 0**
6. Action: **Send email** or **Run a script**

9. Quick Reference: File Locations

File	Location	Purpose
Splunk license config	/opt/splunk/etc/system/local/server.conf	Activate Splunk Free license
Syslog config	/etc/syslog.conf	Forward logs to Splunk
Rockyou wordlist	/usr/share/wordlists/rockyou.txt	Password list for Hydra
Threat intel lookup	/opt/splunk/etc/apps/search/lookups/threat_intel.csv	Simulated OpenCTI enrichment
Splunk binary	/opt/splunk/bin/splunk	Splunk CLI management

10. MITRE ATT&CK Technique Mapping

Technique ID	Technique Name	Lab Activity
T1046	Network Service Scanning	Nmap ACK scan against Metasploitable2
T1110	Brute Force	Hydra FTP brute force against msfadmin
T1078	Valid Accounts (attempted)	Repeated login attempts with msfadmin
T1021	Remote Services	FTP port 21 used as attack vector

11. D3FEND Countermeasure Mapping

ATT&CK Technique	D3FEND Countermeasure	Status
T1046	Network Traffic Filtering	Partial
T1110	Credential Hardening	Gap
T1078	Identifier Activity Analysis	Covered
T1021	Network Traffic Filtering	Gap

End of Source Code Reference

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